Extracting the Born cross section for $ep \to e'p'J/\psi$ at CLAS12: procedure, master formula, and the traps in our own samples

Campaign `prod_jpsi_f18in_20261009`

`exclurad_py` note, branch `jpsi-born-xsec-note`

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Abstract

What to do with a data yield binned in (W, Q^2, t), a radiative Monte Carlo and a Born Monte Carlo in order to obtain a Born cross section. The master formula is given in two algebraically equivalent forms. The one to *compute* is the ratio of the data yield to the reconstructed radiative Monte Carlo, times the model: in that form a wrong model *normalisation* cancels exactly, which matters here because `jpsi.diffrad_vm` sits a factor 4–6 below the data. Only the shape inside the bin survives, and it is iterated away. Six inputs beyond the three samples are required and are listed explicitly; the branching ratio $\mathcal{B}(J/\psi \to e^+e^-)$ is one of them and is *not* applied anywhere in the generator, a factor 16.7 if missed. Moreover the PDG branching ratio is the *radiation-inclusive* leptonic width, so radiation of the decay leptons belongs to the final state that $\mathcal{B}$ already counts and must not be allowed into C^{rad}: the Born arm has to be the `borndecay` one, with Born production and decay radiation on (Sec. 5.2). That premise is checked against the PDG listing, which quotes the radiative submode separately for $E_\gamma > 100$ MeV: the generator's decay model gives $14.38 \pm 0.06\,\%$ against the PDG's $14.74 \pm 2.35\,\%$, agreeing to 0.15σ (Sec. 5.3). Several properties of the existing campaigns were verified while writing this note and are reported in Sec. 5, including the card fields of all three arms and the fact that the generator's `sigma_nb` cannot be used to form the radiative correction.

1 The inputs

Three samples are not enough. The complete set is Table 1. Items 4–10 are not refinements: each is a factor, or a bias of the order of the effect being measured.

#	Input	Source
1	data yield in (W, Q^2, t) bins	analysis ntuple
2	radiative MC through GEMC	`osg_mariana_lund`, arm `rad`
3	**borndecay** MC through GEMC	`osg_mariana_lund`, arm `borndecay`; Sec. 5.2
4	integrated luminosity $\mathcal{L}$	for exactly the run/trigger selection that made the yield
5	signal extraction prescription	$M(ee)$ window and background shape
6	$\mathcal{B}(J/\psi \to e^+e^-) = (5.971 \pm 0.032)\,\%$	PDG. **Not in the generator**, Sec. 5.1
7	data/MC efficiency scale factors	FT electron, lepton and proton tracking, trigger
8	bin volumes, centroids, averaging rule	chosen once and written down
9	one cut list for data and *both* MC arms	exclusivity cut included
10	a model to iterate with	Sec. 3, step 7

Table 1: Inputs to the extraction.

2 The master formula

Each arm carries a per-event weight built from what the generator already reports in `<job>.config.json`,

$$w^{\mathrm{arm}} = \frac{\sigma^{\mathrm{arm}}_{\mathrm{gen}}}{N^{\mathrm{arm}}_{\mathrm{gen}}} \quad [\mathrm{nb/event}], \qquad \sigma_{\mathrm{gen}} \equiv \texttt{report.sigma_nb}. \tag{1}$$

Let n_i be the number of *reconstructed* Monte Carlo events of a given arm falling in bin i after the identical analysis chain, and m_i the number of *generated* (truth) events in bin i with no cuts at all.

Form A — ratio to the radiative MC. This is the one to compute. R_i is a ratio of two *radiative-level* numbers: numerator and denominator are both reconstructed, after cuts, with radiation present, because the data are radiative and the comparison must be like with like. There is no Born quantity in R_i at all — the only Born object in Form A is the model factor multiplying it.

$$\boxed{\sigma_i^{\mathrm{Born}} = R_i\, \bar{\sigma}_i^{\mathrm{Born,model}}, \qquad R_i = \frac{N_i^{\mathrm{data}}}{\mathcal{L}\,\mathcal{B}\, w^{\mathrm{rad}}\, n_i^{\mathrm{rad}}}} \tag{2}$$

Form B — explicit factors. This is the one to tabulate systematics on.

$$\bar{\sigma}_i^{\mathrm{Born}} = \frac{N_i^{\mathrm{data}}}{\mathcal{L}\,\mathcal{B}\,\Delta\Omega_i\,\varepsilon_i^{\mathrm{Born}}\,C_i^{\mathrm{rad}}}, \qquad \varepsilon_i^{\mathrm{Born}} = \frac{n_i^{\mathrm{bdec}}}{m_i^{\mathrm{bdec}}}, \qquad C_i^{\mathrm{rad}} = \frac{w^{\mathrm{rad}}\, n_i^{\mathrm{rad}}}{w^{\mathrm{bdec}}\, n_i^{\mathrm{bdec}}}. \tag{3}$$

Here `bdec` is the **borndecay** arm: Born production with decay radiation *on*. It is not the `born` arm. Section 5.2 gives the reason, and it is the one choice in this note that cannot be got wrong without spoiling the answer.

What Form A assumes, and what it does not

Write the whole chain — radiation, resolution, migration, acceptance, cuts — as one operator K that takes a *Born* cross section in and returns a *reconstructed radiative* yield. Then

$$N_i^{\mathrm{data}} = \mathcal{L}\,\mathcal{B}\,(K\sigma^{\mathrm{true}}_{\mathrm{Born}})_i, \qquad \mathcal{L}\,\mathcal{B}\, w^{\mathrm{rad}} n_i^{\mathrm{rad}} = \mathcal{L}\,\mathcal{B}\,(K\sigma^{\mathrm{model}}_{\mathrm{Born}})_i, \tag{4}$$

so R_i is the ratio of two *outputs* of K, and it factorises:

$$R_i = \frac{(K\sigma^{\mathrm{true}})_i}{(K\sigma^{\mathrm{model}})_i} = \underbrace{\frac{\sigma^{\mathrm{true}}_{\mathrm{Born}}}{\sigma^{\mathrm{model}}_{\mathrm{Born}}}}_{\text{wanted}} \times \underbrace{\frac{C^{\mathrm{rad}}_{\mathrm{true}}}{C^{\mathrm{rad}}_{\mathrm{model}}}}_{\text{assumed}=1} . \tag{5}$$

R_i is therefore the Born ratio *only if* $C^{\mathrm{rad}}_{\mathrm{true}} = C^{\mathrm{rad}}_{\mathrm{model}}$. That is the assumption of the method, and it is not free: it is the physics content of the RC engine, tested by the two-method comparison of `docs/rc_method_jpsi/` and by the generator-level closure to 0.1 %, not by this formula.

Equivalently, Form A is exact if and only if $\sigma^{\mathrm{true}}/\sigma^{\mathrm{model}}$ is *constant* over the region feeding bin i — the bin plus the migration tails plus the radiative smearing. K is then linear there, $(Kc\sigma)_i = c(K\sigma)_i$, and $R_i = c$ exactly. Step 7 is what makes that hold: after reweighting, the ratio is locally constant by construction. So the overall normalisation of the model cancels exactly, while the model's *shape* inside the bin and the correctness of C^{rad} do not.

Why the two forms are the same, exactly

Substituting ε and $C^{\rm rad}$ into Form B, $n_i^{\rm bdec}$ *cancels*:

$$\varepsilon_i^{\rm Born}\, C_i^{\rm rad} = \frac{n_i^{\rm bdec}}{m_i^{\rm bdec}} \cdot \frac{w^{\rm rad} n_i^{\rm rad}}{w^{\rm bdec} n_i^{\rm bdec}} = \frac{w^{\rm rad} n_i^{\rm rad}}{w^{\rm bdec} m_i^{\rm bdec}}, \qquad \bar{\sigma}_i^{\rm Born} = \frac{N_i^{\rm data}\, w^{\rm bdec} m_i^{\rm bdec}}{\mathcal{L}\,\mathcal{B}\,\Delta\Omega_i\, w^{\rm rad} n_i^{\rm rad}}. \tag{6}$$

Comparing with Form A gives the exact condition for the identity,

$$\boxed{\bar{\sigma}_i^{\rm Born,model} = \frac{w^{\rm bdec}\, m_i^{\rm bdec}}{\Delta\Omega_i}} \tag{7}$$

that is, the model's bin-averaged Born cross section taken from the *truth* counts of the Born Monte Carlo — generated events in the bin, times the cross section per event, per unit bin volume — rather than from an integral of the model. Use Eq. (7) and the two forms agree numerically, not merely algebraically. That is what "consistently" means.

Form A is what should be computed, for three reasons.

1. Acceptance, efficiency, resolution smearing, bin migration and radiation all sit inside one measured number, $n_i^{\rm rad}$, obtained by running the same analysis code over reconstructed Monte Carlo. Nothing is factorised that does not factorise.

2. The $M(ee)$ window efficiency cancels. The same window is applied to the data and to $n_i^{\rm rad}$, so it never has to be computed, and the radiative lineshape is simulated rather than assumed.

3. **A wrong model normalisation cancels exactly.** If the model is scaled by a factor f, then $n_i^{\rm rad} \to f\, n_i^{\rm rad}$, hence $R_i \to R_i/f$ and $\bar{\sigma}_i^{\rm Born,model} \to f\, \bar{\sigma}_i^{\rm Born,model}$, so

$$\sigma_i^{\rm Born} = R_i\, \bar{\sigma}_i^{\rm Born,model} \quad \text{is invariant under} \quad \sigma^{\rm model} \to f\, \sigma^{\rm model}. \tag{8}$$

The third point is not academic. The model `jpsi.diffrad_vm` carries $\sigma_T \propto Q^{-3.11}$, which double-counts the $1/Q^2$ already present in the virtual-photon flux, and it lies a factor 4–6 below the data at Forward Tagger kinematics. Form A is immune to that. What survives is only the *shape inside the bin*, and step 7 iterates it away.

3 Procedure

Step 1 — one cut list. Write the cuts once and apply them to the data, the radiative MC and the Born MC from the same code. Include the exclusivity cut. Both cards carry `vcut = 0`, so nothing was cut at generation and the whole radiative phase space is in the files; the exclusivity cut is applied *offline*, and that is the only way migration through resolution is simulated.

Step 2 — signal yield per bin. Count in a window around $M(ee) = 3.0969$ GeV. Do *not* fit a free lineshape: a free Gaussian width absorbs the radiative tail and biases the yield. Window counting reduces that model bias by about a factor 5. Subtract background from sidebands or a fitted continuum, and propagate the subtraction error into $N_i^{\rm data}$.

Step 3 — the same window on the MC. The identical $M(ee)$ window applied to the reconstructed radiative MC gives $n_i^{\rm rad}$. Both arms used here run `idecay = 2`, so the decay final-state radiation is simulated and the lineshape is never assumed. That is what makes the window efficiency cancel between the data and $n_i^{\rm rad}$, and what keeps the decay radiation *out* of $C^{\rm rad}$, where it would be double-counted against $\mathcal{B}$ (Sec. 5.2).

Step 4 — normalise. $Y_i^{\rm rad} = \mathcal{L}\,\mathcal{B}\,w^{\rm rad}\,n_i^{\rm rad}$. Average `sigma_nb` over jobs; all production jobs share the card, so $w^{\rm rad}$ is common to them.

Step 5 — form R_i and multiply by the model. Bin-average the model's Born cross section over the bin using the model's own density, *or* quote at the bin centroid with a bin-centring correction. State which. The two differ wherever the t-slope is steep, which near threshold is everywhere.

Step 6 — compute $C_i^{\rm rad}$ in any case. Form A does not need it, but it must be reported, and its bin-to-bin behaviour is the diagnostic that says whether the correction is sane. A $C^{\rm rad}$ that is flat in W, or above unity, means something upstream is wrong. The published 0.80/0.76/0.62 by W bin was measured with the `born` arm in the denominator, so it carries the decay-lepton factor as well; the extraction's $C^{\rm rad}$ = rad/borndecay will sit closer to unity.

The third arm is *not* a free check on $F(K)$, as an earlier revision of this note claimed. Measured over 300 jobs per arm, borndecay/born $= 0.98879 \pm 0.01296$. Since $\int MB$ is identical for the two arms and `written` = 5000 in both, that ratio is purely the effect of decay FSR on the *generation* daughter window ($|p| \in [1, 9]$ GeV, $\theta \in [5, 40]^\circ$). The analysis cut $|p| > 1.95$ GeV is tighter, so it is $F(K)$ for a window the analysis does not use, and at $1.1 \pm 1.3\,\%$ it is consistent with nothing happening. The decay-radiation model is tested against the PDG instead, Sec. 5.3.

Step 7 — iterate. Reweight the Monte Carlo event by event by $\sigma^{\rm extracted}/\sigma^{\rm model}$ as a function of (W, Q^2, t), then recompute $n_i^{\rm rad}$ and R_i. Two or three passes converge. This is required rather than cosmetic: the model is fitted over $Q^2 = 0.015$–0.1 GeV^2 while the generation box runs 0.001–0.25, so without iteration the bin-averaging is model-dominated at both Q^2 edges.

Step 8 — efficiency scale factors. GEMC does not reproduce the Forward Tagger electron or the tracking efficiencies. Apply the measured data/MC ratios to $Y_i^{\rm rad}$, per bin wherever they are kinematics-dependent.

Step 9 — quote it. For $ep \to e'p'J/\psi$ no flux factor is needed. If converting to $\sigma(\gamma^* p \to J/\psi\, p)$, state the flux convention (Hand or Gilman) explicitly: it moves the number.

4 Statistical error

From Form A, with the three counts independent,

$$\left(\frac{\delta R_i}{R_i}\right)^2 = \frac{1}{N_i^{\rm data}} + \frac{1}{n_i^{\rm rad}} + (\text{background subtraction term})\,. \tag{9}$$

The `borndecay` arm enters the statistical error *only* through $C^{\rm rad}$ and $\varepsilon^{\rm Born}$, that is only in Form B and in the reported diagnostics. That is not a reason to generate less of it.

5 Facts checked in the actual samples, 6 November 2026

5.1 The branching ratio is not in the generator

```
grep -rniE "branching|0.0597|5.97" --include=*.py exclurad_py/   ->   nothing
```

`report.sigma_nb` is the **full J/ψ production cross section**. Every event is decayed to e^+e^- with probability one (`idecay = 2`). The data yield is an e^+e^- yield. Therefore $\mathcal{B}$ multiplies the Monte Carlo, once. Omitting it makes the answer wrong by 16.7.

5.2 The Born arm must be `borndecay`, because $\mathcal{B}$ is the full width

This is the crux of the whole procedure (VPK, 6 November 2026).

The normalisation uses $\mathcal{B}(J/\psi \to e^+e^-) = (5.971 \pm 0.032)\,\%$ from the PDG. That number is the *radiation-inclusive* leptonic width: $\Gamma(J/\psi \to e^+e^-)$ is measured inclusively, so $J/\psi \to e^+e^-\gamma$ is already counted inside it.

Radiation of the decay leptons is therefore *not* a correction to be removed. It is part of the physical final state that $\mathcal{B}$ counts. Its only effect on the measurement is on the *acceptance* — through the $M(ee)$ window and through the lepton momenta — so it must be simulated in every arm that is used, so that it cancels in $C^{\rm rad}$ and is folded correctly into ε. Put it into $C^{\rm rad}$ instead and it is counted twice: once in $\mathcal{B}$, once in the correction.

The Born arm for an extraction is therefore the one with Born *production* and decay radiation *on*. The `osg_mariana_lund` campaign carries exactly that, as a third arm. Card fields verified on ifarm, 6 November 2026:

field	switches	`rad`	`born`	`borndecay`
1	model	`jpsi.vmd2026_pdet`		
4	`iborn` — vertex radiation	0	1	1
8	`vcut`	0.0	0.0	0.0
12	`idecay` — decay-lepton radiation	2	1	2
18	`itarg` — pre-vertex radiator	1	0	0

$$C^{\rm rad} = \frac{\rm rad}{\rm borndecay}: \quad \text{production radiation only (vertex + pre-vertex straggling), decay radiation cancels.} \tag{10}$$

Statistics: 2000 LUND files of 5000 events per arm per period (fa18 / sp18 / sp19), and all three arms are linked into each of the nine OSG dataset directories, so `born`, `borndecay` and `rad` are all submitted.

The `born` arm (`idecay` = 1) is *not* wrong. It was built for the RC-method comparison of `docs/rc_method_jpsi/`, where method 2 multiplies the analytic decay-lepton factor $F(K)$ explicitly, so keeping the decay radiation inside $C^{\rm rad}$ was deliberate and consistent there. It is the wrong denominator for a cross-section extraction, which is a different question.

The model differs between the two campaigns: `osg_mariana_lund` uses `jpsi.vmd2026_pdet` while `prod_jpsi_f18in_20261009` uses `jpsi.diffrad_vm_pdet`. Form A is invariant under the model normalisation, so that costs nothing there; only the shape inside the bin matters, and step 7 iterates it away.

5.3 The decay-radiation model agrees with the PDG radiative branching fraction

The premise of Sec. 5.2 — that $\mathcal{B}$ is radiation-inclusive — is in the PDG listing itself, and the listing also supplies a number the generator can be tested against. PDG 2025, $J/\psi(1S)$, with $e^+e^-\gamma$ *indented* under e^+e^-, which in PDG notation marks a submode, i.e. a subset of the line above:

	mode		branching fraction	
Γ_5	e^+e^-		(5.971 ± 0.032)	%
Γ_6	$\quad e^+e^-\gamma$	[a]	(8.8 ± 1.4)	$\times 10^{-3}$
Γ_7	$\mu^+\mu^-$		(5.961 ± 0.033)	%

[a] For $E_\gamma > 100$ MeV.

So $\mathcal{B}(e^+e^-)$ is the full width and the radiative part is quoted as a fraction of it:

$$\frac{\Gamma_6}{\Gamma_5} = \frac{8.8 \times 10^{-3}}{5.971 \times 10^{-2}} = 14.74 \pm 2.35\,\% \qquad \text{for } E_\gamma > 100 \text{ MeV}. \tag{11}$$

Since $M_{ee}^2 = M^2 - 2Mk^*$, that threshold is $M(ee) < 2.99523$ GeV. Calling our own decay model directly, with no production and no acceptance cut:

`DecayFSR` (`idecay` = 2, exact ME), 300 000 decays	fraction
$E_\gamma > 20$ MeV	23.95 %
$E_\gamma > 50$ MeV	18.63 %
$E_\gamma > 100$ MeV	**14.38 ± 0.06 %**
$E_\gamma > 200$ MeV	10.12 %
$E_\gamma > 400$ MeV	6.06 %
$E_\gamma > 800$ MeV	2.56 %
PDG, same threshold	**14.74 ± 2.35 %**
difference	$-0.36\,\% = 0.15\sigma$

Mean $E_\gamma = 72.0$ MeV, maximum 1548 MeV. This is the first independent check on `generator/decay_rad.py`. It also explains the card's own remark that "$\sim$15 % of decays end below $M(ee) = 3.0$ GeV": 3.0 GeV corresponds to $k^* = 95.4$ MeV, essentially the PDG threshold, which is why the two numbers coincide.

5.4 The card header of `prod_jpsi_f18in_20261009` is false

That campaign's radiative card states "Born twin: IDENTICAL except iborn (field 4)" and offers a `diff` command to prove it. The command prints five differences: fields 4, 11 (seed), 12 (`idecay`), 13 (`outroot`) and 18 (`itarg`). The danger is not the confusion but the repair: a reader who trusts the comment may "fix" field 12 or field 18 to make the diff clean, and a Born arm whose `itarg` or `idecay` has been flipped silently changes what $C^{\rm rad}$ means. It is the comment that is wrong, not the cards.

That campaign's Born arm also has `idecay = 1`, which is why it is not the arm to use for an extraction (Sec. 5.2). Its `itarg = 0` is nevertheless safe from the point-vertex bug: the fix `bbe6628` (2026-10-22 11:25) is an ancestor of the sample's recorded commit `8ac2fc0` (2026-10-22 12:36), so v_z is spread over the 5 cm cell rather than pinned to zero.

5.5 $C^{\rm rad}$ cannot be taken from `sigma_nb`

arm	`report.sigma_nb`
born	3.494470×10^{-4}
rad	3.368744×10^{-4}
ratio rad/born	0.9640

`sigma_nb` counts events *before* the acceptance preset, so any rad/Born ratio built from it is about 1.33 % high, and in any case it is a box average rather than a bin correction. $C_i^{\rm rad}$ must come from reconstructed yields inside the bin.

5.6 The `prod_jpsi_f18in_20261009` arms are not at equal statistics

	born	rad
`out_inb`	1120	1167
`out_outb`	852	1180

The Born outbending arm is 28 % short, which caps the precision of both $C^{\rm rad}$ and $\varepsilon^{\rm Born}$. It should be finished before a systematic is quoted on either.

5.7 `iacc = jpsi` is a hard generation cut, not a flag

`events.py` executes `continue` on failure: the event is never written. The windows travel with the sample in the `.lund.info` sidecar:

particle	$\lvert p\rvert$ (GeV)	θ (deg)
proton	0.2–4.5	3–45
daughter	1.0–9.0	5–40
electron	0.1–2.5	2–20

If any analysis cut is *looser* than the corresponding preset edge, the Monte Carlo has exactly zero events there and the acceptance is wrong, not merely imprecise. Measured in a 2000-event sample, the live edges are the proton θ floor (minimum 3.013°, zero events below) and the daughter θ ceiling (maximum 39.96°). The proton $|p|$ window never bites: the kinematic minimum is 0.7156 GeV/c against a 0.2 GeV/c floor.

5.8 t from the proton, and the choice of t or t'

The proton t estimator has RMS 0.047 GeV^2 against 0.230 for the J/ψ one, a factor 5–6 better. The existing analysis ntuple carries `t_prime`, not `t`. Since $|t|_{\min}$ runs from 2.10 GeV^2 at threshold to 0.4536 at the top of the W range, bins in t' and in t map onto appreciably different regions. Fix the choice once and use it in the data, in both Monte Carlo arms, and in the model bin-averaging.

For reference, the recoil proton follows from t alone through $t = (p - p')^2 = 2M^2 - 2ME_p$, so

$$E_p = M + \frac{|t|}{2M}, \qquad |\mathbf{p}_p| = \sqrt{|t|}\,\sqrt{1 + \frac{|t|}{4M^2}}. \tag{12}$$

At the kinematic ceiling $W^2 = M^2 + 2ME_{\text{beam}} = 20.7717$ GeV^2 ($W = 4.5576$ GeV),

$$|t|_{\min} = 0.4536 \text{ GeV}^2, \qquad |\mathbf{p}_p| = 0.7156 \text{ GeV}/c, \qquad \theta_p = 0.67^\circ, \tag{13}$$

so at the forward edge the recoil proton travels down the beamline and is undetectable; the 3° preset floor removes it. Over the whole box the recoil proton never exceeds $\theta_p = 25.3^\circ$: it is a Forward Detector particle everywhere, never a Central Detector one.

6 Systematics

Source	How to estimate
model shape inside the bin	spread over iterations; vary the Q^2 slope within its fit error
exclusivity cut	vary the MX^2 window; C^{rad} is the quantity that moves
signal extraction	window width; sideband versus fitted background
MC statistics	n^{rad} and n^{bdec} separately
efficiency scale factors	their own uncertainties, bin dependent
luminosity	from the data side
branching ratio	0.54 % relative
bin centring	centroid versus bin-averaged model

Table 2: Systematic uncertainties to carry through Eq. (3).

7 Ntuple branches the chain needs

The reader in `production/analysis/campaign_plots.py` uses

```
E_photon, Q2, W, t_prime, invariantMass,
scattered_theta, scattered_p, proton_theta, proton_p,
positron_theta, positron_p, electron_theta, electron_p,
time_ee, x_FT, y_FT
```

The data and every Monte Carlo arm must expose the same set, filled by the same code, or step 1 cannot be honoured.

Companion notes

- `docs/RC_METHOD_FOR_JPSI.md` — how the π^0 RC-method study was done and what to change for J/ψ.
- `docs/RC_METHOD_COMPARISON.md` — MC ratio versus the analytic `rc_factor`, and what the difference costs.
- `docs/rc_method_jpsi/rc_method_jpsi_report.tex` — the measured C^{rad} for this campaign, both methods.
- `docs/BORN_XSEC_EXTRACTION_JPSI.md` — the Markdown form of this note.